

Weekly Reward Draw System

Complete Documentation — Setup to Production

Version: 1.0
Framework: Laravel 12
Database: MySQL
Frontend: Blade + Tailwind CSS
Real-time: Laravel Reverb (WebSockets)

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1. System Overview

Weekly Reward Draw System is a web application that manages three weekly reward pools:

Pool	Entry Fee	Draw Frequency
Bronze	10 PKR	Every Friday at 00:00
Silver	50 PKR	Every Friday at 00:00
Gold	100 PKR	Every Friday at 00:00

Core features

- User registration, login, and **email OTP verification**
- Manual payment submission (transaction ID + screenshot)
- Admin approval workflow for entries
- **Weekly automated draw** — 1 winner per pool
- **Weekly reset** — soft-delete entries, reopen pools, notify all users
- **Real-time updates** via WebSockets (admin + user dashboards)
- Mobile-first responsive UI (Tailwind CSS)

High-level flow

```
User registers → OTP email verification → Join pool → Submit payment proof
  → Admin approves entry → Friday draw selects winner → Emails sent
  → Entries soft-deleted → Pools reopen → Users re-enter for new week
```

2. Requirements

Server requirements

Component	Minimum Version
PHP	8.2+
MySQL	5.7+ / MariaDB 10.3+
Composer	2.x
Node.js	18+
npm	9+

PHP extensions

- BCMath, CType, cURL, DOM, Fileinfo, JSON, Mbstring, OpenSSL, PCRE, PDO, Tokenizer, XML

Optional (production)

- Supervisor (for Reverb + scheduler)
 - SMTP mail server
 - SSL certificate (HTTPS)
-

3. Installation (XAMPP / Windows)

Step 1: Clone or copy project

Place the project in your web root:

```
C:\xampp\htdocs\lottery-system
```

Step 2: Install PHP dependencies

```
cd C:\xampp\htdocs\lottery-system
composer install
```

Step 3: Install Node dependencies & build assets

```
npm install
npm run build
```

For development with hot reload:

```
npm run dev
```

Step 4: Environment file

```
copy .env.example .env
php artisan key:generate
```

Edit `.env` with your database and mail settings (see [Section 4](#)).

Step 5: Create MySQL database

Start **MySQL** in XAMPP Control Panel, then:

```
mysql -u root -e "CREATE DATABASE lottery_system CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;"
```

Or create `lottery_system` via phpMyAdmin.

Step 6: Run migrations and seeders

```
php artisan migrate --seed
```

This creates all tables and seeds:

- Bronze, Silver, Gold pools
- Default admin account

Step 7: Storage link

```
php artisan storage:link
```

Payment screenshots are stored in `storage/app/public/payments` and served via `public/storage`.

Step 8: Fix soft-deletes column (if needed)

If you see Unknown column 'entries.deleted_at' :

```
php artisan entries:repair-soft-deletes
php artisan migrate
```

4. Environment Configuration

Application

```
APP_NAME="Weekly Reward Draw"
APP_ENV=local
APP_DEBUG=true
APP_URL=http://localhost:8000
```

For XAMPP without `artisan serve` :

```
APP_URL=http://localhost/lottery-system/public
```

Database (MySQL)

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=lottery_system
DB_USERNAME=root
DB_PASSWORD=
```

Mail (SMTP example — Gmail)

```
MAIL_MAILER=smtp
MAIL_HOST=smtp.gmail.com
MAIL_PORT=587
MAIL_USERNAME=your-email@gmail.com
MAIL_PASSWORD=your-app-password
MAIL_ENCRYPTION=tls
MAIL_FROM_ADDRESS=your-email@gmail.com
MAIL_FROM_NAME="Weekly Reward Draw"
```

For local testing without SMTP:

```
MAIL_MAILER=log
```

Emails are written to `storage/logs/laravel.log` .

Broadcasting / WebSockets (Reverb)

```
BROADCAST_CONNECTION=reverb
QUEUE_CONNECTION=sync

REVERB_APP_ID=your-app-id
REVERB_APP_KEY=your-app-key
REVERB_APP_SECRET=your-app-secret
REVERB_HOST=localhost
REVERB_PORT=8080
REVERB_SCHEME=http
```

```
VITE_REVERB_APP_KEY="{{REVERB_APP_KEY}}"
VITE_REVERB_HOST="{{REVERB_HOST}}"
VITE_REVERB_PORT="{{REVERB_PORT}}"
VITE_REVERB_SCHEME="{{REVERB_SCHEME}}"
```

Generate Reverb keys:

```
php artisan tinker --execute="echo Str::random(20);"
```

5. Running the Application

Option A: Development (recommended)

Run all services at once:

```
composer run dev
```

This starts:

- Laravel server (`php artisan serve`)
- Reverb WebSocket server
- Log viewer (Pail)
- Vite dev server

Option B: Manual (separate terminals)

Terminal 1 — Web server:

```
php artisan serve
```

App URL: `http://127.0.0.1:8000`

Terminal 2 — WebSocket server:

```
php artisan reverb:start
```

Terminal 3 — Frontend (development only):

```
npm run dev
```

Option C: XAMPP Apache

1. Start Apache + MySQL in XAMPP
2. Start Reverb: `php artisan reverb:start`
3. Open: `http://localhost/lottery-system/public`

Ensure `mod_rewrite` is enabled and `.htaccess` in `public/` is active.

Connection indicator

- **Green dot** in header = WebSocket connected

- **Red dot** = Reverb not running
-

6. Default Credentials

Role	URL	Email	Password
Admin	/admin/login	admin@rewarddraw.test	password
User	/register (create new)	—	—

Change the admin password immediately in production.

7. User Guide

7.1 Registration

1. Go to `/register`
2. Enter name, email, password
3. After registration you are redirected to **OTP verification**

7.2 Email OTP verification

- A **6-digit OTP** is sent to your email
- OTP expires in **10 minutes**
- Resend is available after **60 seconds** cooldown
- You must verify before accessing pools

7.3 Dashboard

After verification, `/dashboard` shows:

- **Bronze / Silver / Gold** pool cards
- Entry fee and participant count
- Countdown to next Friday draw
- **New Week Started** banner (after weekly reset)
- **Verified User** badge

7.4 Joining a pool

1. Click **Join Pool** on an open pool
2. Transfer the exact entry fee (10 / 50 / 100 PKR) via your payment method
3. Submit:
 - **Transaction ID** (must be unique)
 - **Payment screenshot** (JPEG/PNG/WebP, max 5MB)
4. Entry status becomes **Pending** until admin approval

7.5 Entry rules

- One entry per pool per week
 - Only **approved** entries participate in the draw
 - Entries do **not** carry over — re-enter every week after reset
 - Email verification is mandatory
-

8. Admin Guide

Admin panel URL: `/admin/login`

8.1 Dashboard

Shows live stats:

- Total users
- Pending / approved entries
- Total winners
- Recent entries table (updates in real-time)

8.2 Manage entries

Path: `/admin/entries`

- View all entries with status filters
- Click **Review** to see payment screenshot
- **Approve** — user becomes eligible for draw
- **Reject** — entry excluded from draw

8.3 Manage pools

Path: `/admin/pools`

- View entry counts per pool
- **Open / Close** pools manually

8.4 Users

Path: `/admin/users`

- List all registered users
- Verification status and entry count

8.5 Winners history

Path: `/admin/winners`

- All past draw winners by pool and week

8.6 Weekly draw

Path: `/admin/draw`

Run Draw & Reset Manually executes the full weekly flow:

1. Lock all pools
2. Select 1 winner per pool (approved + verified users only)
3. Send winner emails
4. Soft-delete all entries for the week
5. Reopen pools
6. Send reset email to all verified users

9. Weekly Draw & Reset System

Automated schedule

Every **Friday at 00:00** (app timezone):

```
php artisan draw:weekly
```

Registered in `routes/console.php` . Requires cron:

```
* * * * * cd /path/to/lottery-system && php artisan schedule:run >> /dev/null 2>&1
```

Manual draw

```
php artisan draw:weekly
```

Optional week override:

```
php artisan draw:weekly --week=202627
```

Winner selection rules

For each pool:

- Only **approved** entries
- Only **email-verified** users
- Random selection of **1 winner**
- Stored in `winners` table with `week_number`

Weekly reset

After draw:

- All entries for that week are **soft-deleted** (`deleted_at` set)
- Pools reopen for new entries
- Users must re-enter — no carry-over

Emails sent

Email	Recipients	Subject
Winner notification	Winners only	Congratulations! You Won...
Weekly reset	All verified users	New Weekly Draw Started - Join Now

10. Real-Time Updates (WebSockets)

Powered by **Laravel Reverb** + **Laravel Echo**.

Events that trigger live updates

Action	User dashboard	Admin panel
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Entry submitted	Status + toast	New row + stats
Entry approved/rejected	Status toast	Stats + row update
Pool toggled	Pool badges	Toast
Draw completed	Pools reset	Stats refresh

Channels

Channel	Type	Audience
pools	Public	Everyone
admin.dashboard	Private	Admins only
user.{id}	Private	Specific user

Start Reverb

```
php artisan reverb:start
```

Production: run Reverb under Supervisor (see [Section 12](#)).

11. Email System

OTP email

- Sent on registration and OTP resend
- Template: `resources/views/emails/otp.blade.php`
- 6-digit code, 10-minute expiry

Winner email

- Sent after weekly draw to winners
- Template: `resources/views/emails/winner.blade.php`
- Includes pool name, prize, week number

Weekly reset email

- Sent to all verified users after draw
- Template: `resources/views/emails/weekly-reset.blade.php`
- Lists last week's winners + join CTA

Debug emails locally

```
MAIL_MAILER=log
```

Check `storage/logs/laravel.log` for email content.

12. Production Deployment

Checklist

- ☐ Set `APP_ENV=production` and `APP_DEBUG=false`
- ☐ Run `php artisan config:cache`, `route:cache`, `view:cache`
- ☐ Run `npm run build`
- ☐ Configure real SMTP mail
- ☐ Change default admin password
- ☐ Set up SSL (HTTPS) — update `REVERB_SCHEME=https`
- ☐ Point web server document root to `public/`
- ☐ Set up cron for scheduler
- ☐ Run Reverb under Supervisor
- ☐ Set correct `APP_URL`

Apache virtual host example

```
<VirtualHost *:80>
    ServerName rewarddraw.example.com
    DocumentRoot "C:/xampp/htdocs/lottery-system/public"

    <Directory "C:/xampp/htdocs/lottery-system/public">
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

Cron (Windows Task Scheduler)

Program: `C:\xampp\php\php.exe`

Arguments: `C:\xampp\htdocs\lottery-system\artisan schedule:run`

Trigger: Every 1 minute

Supervisor — Reverb (Linux)

```
[program:reverb]
command=php /var/www/lottery-system/artisan reverb:start
autostart=true
autorestart=true
user=www-data
```

File permissions

Ensure writable:

- `storage/`
- `bootstrap/cache/`

13. Troubleshooting

Unknown column 'entries.deleted_at'

```
php artisan entries:repair-soft-deletes
php artisan migrate
php artisan config:clear
```

WebSocket not connecting (red dot)

1. Ensure Reverb is running: `php artisan reverb:start`
2. Check `.env` Reverb variables match `VITE_REVERB_*`
3. Rebuild assets: `npm run build`
4. Clear config: `php artisan config:clear`

Emails not sending

1. Verify SMTP credentials in `.env`
2. For Gmail, use an **App Password** (not your login password)
3. Test with `MAIL_MAILER=log` first

Duplicate transaction ID error

Each transaction ID must be unique across active (non-deleted) entries.

Migration fails on unique index drop

Run:

```
php artisan migrate
```

The repair migration handles partial states automatically.

Nothing to migrate but column missing

```
php artisan entries:repair-soft-deletes
```

Draw selects no winners

- Ensure entries are **approved**
- Users must be **email verified**
- Check entries exist for current `week_number`

14. Export to PDF

Option A: Browser print

1. Open `docs/WEEKLY-REWARD-DRAW-SYSTEM.md` in VS Code or GitHub
2. Use a Markdown preview extension
3. Print → Save as PDF

Option B: Pandoc (if installed)

```
cd docs
pandoc WEEKLY-REWARD-DRAW-SYSTEM.md -o WEEKLY-REWARD-DRAW-SYSTEM.pdf --pdf-engine=xelatex -V
geometry:margin=1in
```

Install Pandoc: <https://pandoc.org/installing.html>

Option C: Online converter

Upload docs/WEEKLY-REWARD-DRAW-SYSTEM.md to any Markdown-to-PDF converter.

15. Project Structure

```
lottery-system/
├─ app/
│   ├─ Console/Commands/
│   │   ├─ WeeklyDrawCommand.php      # Friday draw cron
│   │   └─ RepairEntriesSoftDeletesCommand.php
│   ├─ Events/
│   │   └─ RealtimeUpdate.php          # WebSocket broadcast event
│   ├─ Http/
│   │   ├─ Controllers/
│   │   │   ├─ Auth/                  # Login, register
│   │   │   ├─ Admin/                 # Admin panel
│   │   │   └─ DashboardController.php
│   │   │   └─ EntryController.php
│   │   │   └─ OtpVerificationController.php
│   │   └─ Middleware/
│   ├─ Mail/                          # OTP, winner, reset emails
│   ├─ Models/                        # User, Admin, Pool, Entry, Winner, EmailOtp
│   └─ Services/
│       ├─ DrawService.php             # Winner selection, soft-delete
│       ├─ EmailService.php            # Winner + reset emails
│       ├─ OtpService.php              # OTP generation/verification
│       ├─ RealtimeService.php         # WebSocket broadcasts
│       └─ WeeklyDrawService.php       # Full weekly orchestration
├─ database/migrations/               # Schema definitions
├─ docs/                             # This documentation
├─ resources/
│   ├─ js/                           # Echo + realtime handlers
│   └─ views/                         # Blade templates
├─ routes/
│   ├─ web.php                       # All web routes
│   ├─ channels.php                  # Broadcast channel auth
│   └─ console.php                   # Scheduler
└─ public/                           # Web root (point server here)
```

Database tables

Table	Purpose
-------	---------

users	Registered users
admins	Admin accounts
email_otps	OTP codes (6-digit, 10 min expiry)
pools	Bronze, Silver, Gold pools
entries	Payment submissions (soft-deletes)
winners	Weekly draw results

16. Artisan Commands Reference

Command	Description
<code>php artisan migrate --seed</code>	Run migrations + seed pools & admin
<code>php artisan storage:link</code>	Link public storage for screenshots
<code>php artisan serve</code>	Start development web server
<code>php artisan reverb:start</code>	Start WebSocket server
<code>php artisan draw:weekly</code>	Run weekly draw + reset manually
<code>php artisan schedule:list</code>	View scheduled tasks
<code>php artisan schedule:run</code>	Run scheduler (use in cron)
<code>php artisan entries:repair-soft-deletes</code>	Fix missing deleted_at column
<code>php artisan config:clear</code>	Clear cached configuration
<code>php artisan route:list</code>	List all application routes
<code>composer run dev</code>	Start server + reverb + vite + logs

Route Reference

User routes

Method	URL	Name
GET	/login	login
POST	/login	—
GET	/register	register
POST	/register	—
POST	/logout	logout
GET	/verify-email	otp.show

POST	/verify-email	otp.verify
POST	/verify-email/resend	otp.resend
GET	/dashboard	dashboard
GET	/pools/{pool}/enter	entries.create
POST	/pools/{pool}/enter	entries.store

Admin routes

Method	URL	Name
GET	/admin/login	admin.login
GET	/admin/dashboard	admin.dashboard
GET	/admin/entries	admin.entries.index
PATCH	/admin/entries/{entry}/approve	admin.entries.approve
PATCH	/admin/entries/{entry}/reject	admin.entries.reject
GET	/admin/pools	admin.pools.index
GET	/admin/users	admin.users.index
GET	/admin/winners	admin.winners.index
GET	/admin/draw	admin.draw.index
POST	/admin/draw/run	admin.draw.run